

Self-healing climate control: gap check and resource check

Date: 2026-09-24. Sources: Consensus (6 searches), OpenAlex duplication test and two gap-saturation curves, plus a read of `firmware/terrarium`. Consensus links are listed as given by Consensus; open each paper and record its DOI before citing.

1. Verdict

Each building block has been published on its own: detecting faults in greenhouses, virtual sensors that stand in for a failed humidity sensor, and small AI models running on an ESP32. What has not been published is the closed loop on real hardware. That means a cheap microcontroller that detects a fault, names the failed part and then reconfigures itself to keep the plant climate in range, tested with physically injected faults, with the climate outcome measured against rule-based handling. The only greenhouse paper that reconfigures after a fault did it in simulation. So the paper is worth doing, but the contribution is the integration plus the measured recovery, not any single component.

2. What is already done

Piece	Done by	What they did	What they did not do
Fault detection and identification in greenhouses	Linker 2000, Computers and Electronics in Agriculture	Hybrid physical and neural-network model, sensor and actuator failures, tested on experimental data	No automatic recovery, PC-based, large greenhouse
Same, observer based	Linker 2002, Control Engineering Practice	Experimental greenhouse; ventilation failure found in under 30 min, slow faults in hours	No recovery
Same, small farm unit	Munser 2026, Journal of Process Control	Residual observers on a small vertical-farming unit, faults distinguished during operation	No recovery, observer theory, no public data
Replacing failed sensors with AI predictions	Shekarian 2024, Computers and Electronics in Agriculture	Cloud 1D-CNNs predict greenhouse humidity (RMSE 3.47 % RH) when a sensor fails	Cloud only, sensors only, no actuator faults, no control outcome
Virtual humidity sensor with accommodation	Ponce 2024, Heliyon	Buildings and a museum store; 95 % fault detection and identification	Not plants, no actuators
Virtual sensors for farming	Chourlias 2025, Internet of Things	Field nodes, public dataset, LightGBM virtual sensors	Monitoring only, no control
Fault-tolerant (fault-hiding) greenhouse control	Hameed 2017, Int. J. Automation and Control	Virtual sensors and actuators restore control after sensor or actuator faults	Simulation only
AI anomaly detection trained on an ESP32	Antonini 2023, Sensors	Isolation forest trained on the chip in seconds, industrial pumps	Detection only, not agriculture
Actuator health in commercial smart farms	Choe 2025, Applied Sciences	Field-validated anomaly detection and remaining-life estimation for switches and sensors	Cloud/web platform, maintenance forecasting, no live recovery
Labelled fault datasets	Attarha 2023, Brujin 2016	Sensor-fault datasets, mostly synthetic injection	No actuator faults, no misting enclosure

The recent surveys ([Zou 2023, Sensors](#); [Daurenbayeva 2023, Energies](#)) list recovery and real-world validation as open problems.

3. What is not done

1. **Detect, identify and recover, all on the microcontroller**, in a plant enclosure, with recovery actions

that act on the real climate (virtual sensor takeover, backup actuator switchover, safe mode).

2. **Physically injected actuator faults** in a small misting enclosure: dead mist disc, empty reservoir, stalled fan, lid left open. Existing agriculture datasets inject sensor faults in software.
3. **Measured climate outcome of a fault**: time in the humidity band, and water or plant exposure, with self-healing versus rule-based fail-safes versus no handling.
4. **AI versus plain rules on the same faults**. This is the comparison reviewers will ask for, and nobody reports it for this class of system.

4. Momentum

- On-device fault detection and fault-tolerant control for greenhouse climate: about 10 papers a year, flat to slowly rising. Not crowded.
- TinyML anomaly detection on the ESP32: 2 papers in 2023, 2 in 2025, 7 in 2026. Rising fast, so publish soon.

Both counts come from a broadened lexical query and are only indicative.

5. The study

Working title: Self-healing humidity control on a low-cost microcontroller: on-device fault detection, isolation and recovery in a misted plant enclosure, validated with physically injected faults.

How it works

1. **Predictor.** A small model (a linear autoregressive model first, then a small neural network if it helps) predicts the next humidity reading from recent humidity, temperature, mist state, fan state and light. It is trained on the laptop from 2 to 3 days of normal logs. The weights are exported as a C array to the ESP32, and the model uses a few kilobytes.
2. **Detection.** The residual between prediction and measurement is watched over a sliding window.
3. **Isolation.** A fault-signature table maps residual patterns and actuator states to the failed part.
4. **Recovery.**
5. Humidity sensor failed or stuck: control continues on the predicted humidity (virtual sensor).
6. Humidity mist disc dead: switch to the second mist module, but only if it shares the same air volume.
7. Tank empty, fan stalled or lid open: safe mode plus the buzzer.

Baselines: (a) the current firmware rules, which turn misting off when humidity is unreadable and rely on the float switch, and (b) no fault handling.

Fault list, with current hardware

Fault	How to inject	Why it matters
BME280 disconnected	Pull the I2C lead	Easy case. Current rules stop misting and the plants dry out
BME280 stuck or fouled	Seal it in a small bag, or wet it	Hard case. Readings look valid but are wrong, and rules miss it
Mist disc dead	Unplug the disc lead, drive stays on	Actuator fault invisible to current firmware
Reservoir empty with a failed float	Empty the tank and tape the float up	Two faults at once, the disc would run dry
Exhaust fan stalled	Block the blades	Only detectable through the climate response
Lid open	Open the lid	Environmental fault, not a hardware one
One soil probe failed	Unplug one of the two probes	Uses the redundant pair

Metrics: detection delay, correct isolation rate, false alarms per day on normal data, time in the humidity band during each fault, and mist-minutes wasted.

Output: the paper, plus the labelled dataset (normal operation and injected faults) released openly.

6. Resources

Item	Status	Note
ESP32	Have	Enough memory for the model
BME280, BH1750, 2 soil probes, float, leak sensor	Have	Soil pair gives sensor redundancy
Two mist modules (GPIO25 humidity, GPIO16 watering)	Have	Check that both discs mist into the same air volume. If not, backup switchover is dropped and recovery is virtual sensor plus safe mode only
Exhaust fan	Have	Circulation fan is not fitted (<code>ENABLE_FAN_CIRC 0</code>), so it is left out of the fault list
Rule-based fail-safes in firmware	Have	Serve as baseline (a)
Serial logging every 2 s, laptop tools	Have	Laptop must stay connected during runs
Reference humidity sensor (SHT31 or second BME280)	Buy, about 500 to 700 taka	Mandatory. It scores the virtual sensor when the main one is disabled. It is never used by the controller
Spare mist disc	Buy, cheap	Fresh disc before the fault campaign
Micro-SD module	Optional, about 200 taka	Only if the laptop cannot stay attached for days
Current sensor on actuators	Not needed	Would make actuator faults trivial and weaken the model-based claim

Total new spend is under 1500 taka.

7. Time (two weeks, tight but feasible)

Days	Work
1 to 3	Add the reference sensor. Log normal operation around the clock with the firmware's normal cycles
4	Train the predictor, set thresholds from normal data only, port to the ESP32
5 to 8	Fault campaign: 7 faults, 5 repeats each, under self-healing, rules and no handling. About 20 minutes per run, roughly 30 hours of attended work
9 to 14	Analysis, figures, writing, dataset packaging

8. Risks

- **"Incremental over Linker 2000 and Shekarian 2024."** The answer is recovery on the chip, actuator faults, the measured climate outcome and open data. Say this in the first paragraph of the introduction.
- **The model does not beat simple rules.** Report it honestly. The recovery results still stand, and a clean negative comparison is publishable.
- **Five repeats is small.** State it, report per-run results, and keep the dataset open so others can extend it.

9. Venues

- Computers and Electronics in Agriculture (Q1). It already published Linker 2000 and Shekarian 2024, so it is the natural home.
- Smart Agricultural Technology (Q1 in recent listings). Realistic first target if the first one rejects.
- Sensors or Internet of Things. Fallback.